

AjJay Kannan

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Summary

Data scientist with 3+ years building production ML and data products in supply-chain domain: demand forecasting, supply planning, inventory optimization, and customer analytics. Skilled in Python, SQL, Snowflake, dbt, Power BI, Azure, and AWS owning end-to-end from modeling to deployment and monitoring in Agile/Scrum teams. Drove measurable wins including reduced stockouts, OTIF +16%, and 1,000+ hours/month saved

Skills

Programming Python, SQL, R

Platform AWS, Azure, Snowflake, dbt, Dataiku, MongoDB

DevOps/Infra Docker, Git, CI/CD with Azure Pipelines

Apps/BI Streamlit, Dash, Flask, Bootstrap, Power BI

ML/AI TensorFlow, Pytorch, Prophet, Optimization

LLM Langchain, Llamaindex, Snowflake Cortex

Experience

CHS Inc

Data Scientist — Supply Chain Planning and Execution

Previously: Data Analyst Intern

St. Paul, MN

Jun 2023 – Present

Jan 2023 – Apr 2023

- Created a **product substitution tool** employing **OpenAI LLM** and deployed **Dash webapp** via **Dataiku**, improving search efficiency by **85%** and increasing in on-time-in-full delivery by **>16%**
- Developed and deployed a **Python** based **Mixed Integer Optimization Model** for Inventory Assignment and Blending Optimization using **CBC Solver** to reduce stockout and improve blend quality
- Segmented customers using **RFM analysis** and **ML clustering** to highlight buying pattern and declining sales
- Delivered **Demand Forecasting** using **Prophet** in **Snowpark** and developing a **Streamlit webapp** utilizing **Snowflake Streams** for **Demand Planning**. Created Power BI Reports to enhance Demand Planning workflows
- Integrated demand, inventory and supply data in Snowflake using **dbt**, delivering **60-day visibility** into stockouts, Capacity Constraints and Capacity Utilization KPIs

Texas A&M University

Data Analyst — Student Technician (Part Time)

College Station, TX

Sep 2021 – Jan 2023

- Reduced churn from **35%** to **20%** in the Engineering Honors Program by conducting **Cohort Analysis** in Tableau, applying **K-Means Clustering** for Segmentation, and performing **Hypothesis Testing**

o9 Solutions Inc

Business Operations Specialist Intern

Dallas, TX

Jun 2022 – Aug 2022

- Developed **Python Plugins** for data integration using **ETL** processes in **Apache Spark** and optimized SQL queries to enhance models for Supply Chain Planning and Control Tower Functionalities

Tata Motors Limited

Operations Manager

Previously: Graduate Engineer Trainee

Pune, India

Aug 2019 – Jun 2021

Aug 2018 – Jul 2019

- Led a 50+ member cross-functional team to deliver 13 CKD vehicle projects, **implementing lean and ML-driven tools** while achieving 100% fulfillment, 23% lower stock-out costs, 4% station downtime, and 30% faster claims closure

Projects

LLM Applications Portfolio

- Developed and deployed multiple LLM prototypes: – **LangChain**-based general-AI agent [🔗](#) – **Text-to-SQL agent** using **Snowflake Cortex Analyst** [🔗](#) – **RAG using Snowflake Cortex backend** [🔗](#) for interacting with pesticide product label

Azure Native ReAct RAG Document Summarization [🔗](#)

- Built an Azure-native ReAct RAG pipeline with **LlamaIndex**, **OpenAI**, **CosmosDB**, and **Azure Functions** for automated document summarization
- Deployed a **Flask web app** with **Bootstrap**, integrated Azure Repos, and implemented **CI/CD** using Azure Pipelines

Quantitative Risk Modeling of Supply Chain Risks on US Automotive GDP [↗](#)

- Built **Bayesian Network** models in AgenaRisk to quantify risks to GDP from trade war, pandemic, and chip shortage

Kaggle Competition - Amex Credit Default Prediction [↗](#)

- Built credit default prediction models using **LGBM** and **TensorFlow (Neural Decision Forest)**, evaluated with averaged Gini and Recall, achieving scores of 0.7917 (Forest) and 0.7954 (LGBM) vs. a 0.8097 benchmark

Process Parameter Optimization using Design of Experiments and ANOVA

- Statistically optimized forming to achieve a 65° forming angle for SS430 and 0.9μ surface finish for Inconel

Education

Texas A&M University <i>Master of Science in Industrial Engineering (GPA: 3.89/4.00)</i>	Aug 2021 – May 2023
PSG College of Technology <i>Bachelor of Engineering in Production Engineering (GPA: 8.61/10.00)</i>	Jun 2014 – June 2018